

**PURCHASE DESCRIPTION
FOR
*GEC MD8060 Scale Force Press Manual Upgrade***

1.0 SCOPE:

This purchase description specifies converting the GEC MD8060-00000-10 (MD8060) to manual hydraulic control. The manually controlled MD8060 will be referred to as the “System”. The System will be used to calibrate aircraft and vehicle weighing scales.

2.0 MANUALLY CONTROLLED HYDRAULIC PUMP:

2.1 A new manually controlled hydraulic pump shall replace the MD8060’s hydraulic power unit to control the application of force. The pump shall allow movement of the ram in both up and down motion.

2.2 The System shall maintain the MD8060’s measurement range of 250 to 60,000 lbf and accuracy of 0.025% of Applied Value.

2.3 The pump shall have a two speed design that can be controlled by a single person. Two speed design is defined as under no-load, the pump operates in the high flow first stage for rapid advance. When the load is contacted, the pump automatically shifts to the second stage for building pressure.

2.4 The pump shall be fixtured to the MD8060. Placement of the pump shall be at approximately the same working height of the MD8060.

2.4.1 The pump shall not exceed 30” L x 30” W.

2.5 Ancillary Equipment: Any special equipment, fixtures, tools, standards, hydraulic fluid, etc., required to install and operate the manually controlled hydraulic pump shall be supplied. This includes drill bits and taps.

2.5.1 Hydraulic Fluid supplied with the System shall include a Material Safety Data Sheet.

2.6 The System shall not require software to operate.

3.0 GENERAL REQUIREMENTS:

3.1 Use of Best Commercial Practices: The equipment shall be designed and constructed of materials that meet the requirements of this specification and in accordance with the best commercial practices for industry approved equipment. It shall be designed and manufactured to maximize reliability, maintainability, accuracy of operation, and ease of operation. All parts (which include, but are not limited to, shafts, bearings, regulators, switches, controls) shall have proper clearance and adjustments. They shall work together so that the equipment will supply the rated requirements without unnecessary strains, vibrations, or overheating. They shall be able to withstand the conditions met in shipping, storage, installation, and service. All materials and parts of this equipment shall be capable of withstanding the environmental requirements of this specification.

3.2 Matched and Selected components: These shall be kept to an absolute minimum. Use of specially built, nonstandard components shall be avoided.

3.3 Model and Serial Number Identification: The System shall receive a new Part Number. Part Number, Manufacturer, and Serial Number shall be identified clearly on the item in a location visible to the operator.

3.4 Operating Environment/Storage:

3.4.1 Operating Temperature: The System shall be capable of operating in an ambient temperature of 67 to 79 degrees F without loss of required performance.

3.4.2 Operating Humidity: The System shall be capable of operating in an atmosphere from 20% to 50% relative humidity without loss of required performance.

3.5 Reversibility: This upgrade shall be reversible to restore the MD8060's automated functionality.

4.0 OPERATOR'S MANUAL: A new manual reflecting the new Part Number shall be IAW DD1423s and Data Item descriptions, and shall also include, at a minimum, the following information:

- A. Setup/installation instructions for the manually controlled hydraulic pump.
- B. Operation Instructions.
- C. Routine Maintenance for the manually controlled hydraulic pump.

5.0 INSTALLATION: The system shall be installed/setup by PMEL personnel.

6.0 WARRANTY AND TECHNICAL SUPPORT: Contractor shall provide a three year hardware warranty with technical support.

7.0 ENVIRONMENT, SAFETY, AND OPERATIONAL HEALTH: A combination of the system manual, other documentation, and/or the system itself shall identify potential environment, safety, and operational health hazards and how they are mitigated when the design itself cannot mitigate them. This will be accomplished through stating the hazard and accompanying mitigation through the use of safety devices, use of warnings, or use of special operational procedures to counter such hazards.

8.0 TESTING: This item will be inspected and accepted at destination as stated in the contract. The item will be checked with test equipment with parameters equal to or better than those required to assure that specified performance parameters are met. Test equipment used shall have first echelon certification directly traceable to the International System of Units through the National Institute of Standards Technology (NIST) or other approved equivalent international laboratory. Tests will be performed to assure that the item meets the requirements of this purchase description. Such testing at destination does not relieve the contractor of performing all inspections and quality checks at the point of manufacture as necessary to assure performance as specified herein.

Attachment:
Data Requirements

**Attachment to
Purchase Description
25M-332A-FO
GEC MD8060 Manual Upgrade**

DATA REQUIREMENTS

MANUALS: A complete user and service manual and calibration procedure shall be provided with each unit in contractor format. Manuals shall be on CD/DVD-ROM in Indexed Portable Document Format (iPDF). The manuals shall comply with *Data Item Description* (DID) DI-TMSS-80527D and the *Contract Data Requirements List* (DD Form 1423).